

Documentation: Bengali Fake Review Dataset — Code-Based Workflow

Dataset Source

We use the Bengali Fake Review Dataset hosted on Hugging Face. Due to the dataset license (CC BY-NC-ND 4.0), dataset files must not be uploaded into GitHub. Instead, instructions are provided to download the dataset locally.

Dataset link: <https://huggingface.co/datasets/shawon95/Bengali-Fake-Review-Dataset>

Workflow Summary

- Setup & Imports: Environment prepared in Colab with pandas, scikit-learn, rapidfuzz. Seed fixed (SEED=42).
- Data Loading: Loaded two CSV files (fake.csv, non-fake.csv). Columns: Review (text), Label (0/1).
- Cleaning & Deduplication: Normalized whitespace, removed duplicates using RapidFuzz (threshold=95).
- Augmentation (FAKE only): Custom augmentation with punctuation tweaks, character swaps, and thin space insertion.
- Balancing: Ensured 1:1 balance between FAKE and NON-FAKE by augmenting or sampling.
- Corpus Assembly: Combined balanced data, deduplicated, and shuffled.
- Stratified Split: Train/Validation/Test = 80/10/10 using stratified sampling with seed.
- Saving Outputs: Exported train.csv, val.csv, test.csv, plus prep_meta.json with metadata.

Key Features of the Pipeline

- Reproducible pipeline (fixed seed, metadata JSON).
- Balanced dataset via FAKE augmentation and NON-FAKE sampling.
- Lightweight augmentation (label-preserving).
- Stratified 80/10/10 split ensures equal class distribution.
- Outputs ready-to-train CSVs with metadata for traceability.

Ethical & Licensing Notes

The dataset is under CC BY-NC-ND 4.0 (Non-Commercial, No Derivatives). Do not upload dataset CSVs or augmented outputs to GitHub. Only share code (preprocessing, augmentation, splitting). This pipeline is strictly for research and educational purposes.